

Subjective Mock Test

25/12/2023
Monday-

PHYSICS

1) 3. Concave mirror

2) $m = 3, u = -10 \text{ cm}$

$$m = \frac{-v}{u}$$

$$3 = \frac{+v}{+10}$$

$$v = \boxed{30 \text{ cm}}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{30} + \frac{1}{10} \Rightarrow \frac{1}{f} = \frac{1+3}{30}$$

$$f = \frac{30}{4} \Rightarrow \boxed{7.5 \text{ cm}}$$

$$\underline{3, 30 \text{ cm}, 7.5 \text{ cm}}$$

3) 4. Both 2 and 3.

4) (i) Principal focus of a spherical mirror is a point on the principal axis where parallel incident rays parallel to the axis meet after reflecting.

(ii) The centre of curvature of a spherical mirror is the point on the principal axis that is the centre of the sphere from which the mirror is cut.

5) $m = 1/5, v = 18,$

$$m = \frac{-v}{u} \Rightarrow \frac{1}{5}$$

$$\frac{-18}{u} = \frac{1}{5}$$

$$u = \boxed{-90 \text{ cm}}$$

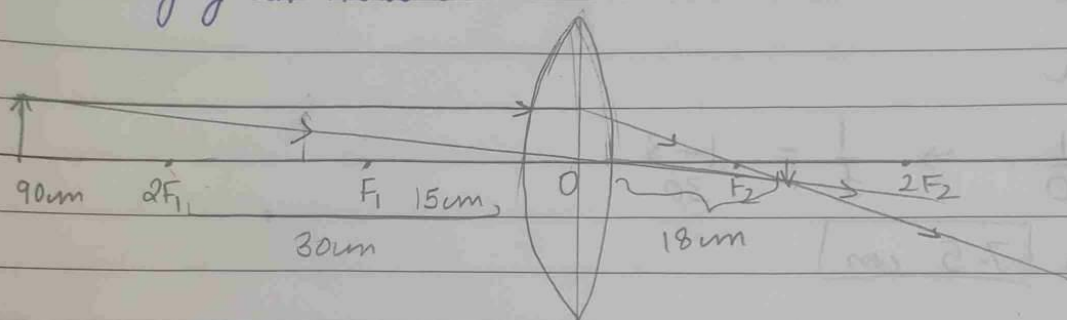
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{18} + \frac{1}{90}$$

$$\frac{1}{f} = \frac{5+1}{90}$$

$$f = \frac{90}{6} \Rightarrow 15 \text{ cm}$$

Since the focal length is +15, the lens is a convex lens, that is converging in nature.



6) (i) $h_o = 7 \text{ cm}$, $u = -27 \text{ cm}$, $f = -18 \text{ cm}$
image formation distance = v

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{-1}{18} = \frac{1}{v} - \frac{1}{27}$$

$$\frac{1}{v} = \frac{1}{27} - \frac{1}{18}$$

$$\frac{1}{v} = \frac{2-3}{54} \Rightarrow -\frac{1}{54}$$

$$v = -54 \text{ cm}$$

The screen should be placed 54 cm in front of the concave mirror.

$$m = \frac{-v}{u} = \frac{+54}{-27} \Rightarrow -2$$

$\therefore$ The image 2 times magnified (enlarged) and is real and inverted.

$$(ii) R = 2f$$

$$f = \frac{5}{2} \text{ m}$$

The mirror is convex

$$u = -20 \text{ m}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{2}{5} = \frac{1}{v} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{2}{5} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{8+1}{20} \Rightarrow \frac{9}{20}$$

$$v = \frac{20}{9} \Rightarrow \boxed{2.22 \text{ m}} \approx \underline{2.22 \text{ m}}$$

the image formed is behind the mirror, 2.22 m.

$$m = \frac{-v}{u} = \frac{-\frac{20}{9}}{-20} \Rightarrow \frac{-20}{9} \times \frac{1}{-20}$$

$$m = \frac{1}{9}$$

The image is virtual, erect and $\frac{1}{9}$ times diminished.

(iii) The mirror is a concave mirror, that is converging in nature. It is used in the instrument used by the dentist as it enlarges the object image, and makes it easier to view the teeth that have to be treated.